

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Pig Trap End Closure
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-010
Document Revision	R01
Document Status	Issued For Review
Originator	Joseph Obinna
Security Classification	Restricted
Issue Date	27 Jul 2022

Data Sheets for Pig Trap End Closure

R01	27.07.22	Issued For Review	J. Obinna	E.Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	27/07/2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-010

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

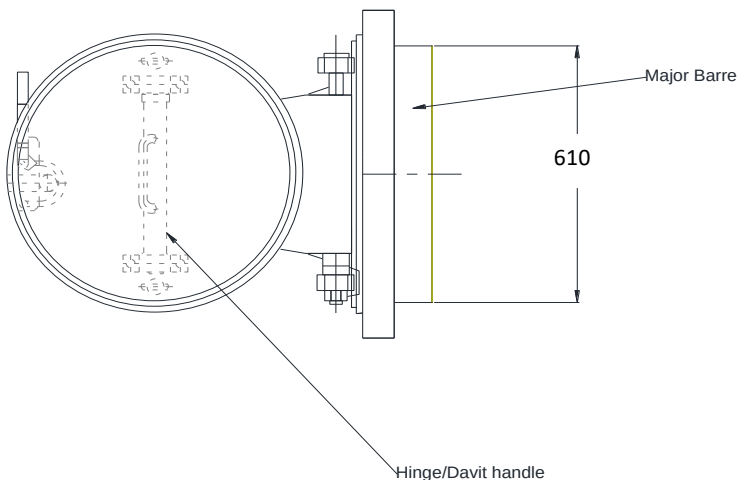
REFERENCES

Document No.	Description

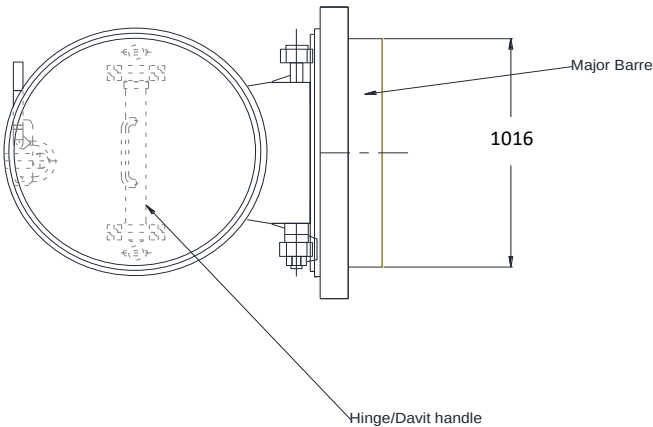
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DATASHEET FOR PIG TRAP END CLOSURE				
Location:		BBOU and ELEPA Manifold Stations (MS)		Tag Numbers: BBOU-A-200, ELEPA-A-200
Equipment / item:		END CLOSURE (QUICK OPENING)		
1	GENERAL	Item	Quick Opening Closure (Note 1, 2)	
2		Equipment attached	Pig Launcher (28"x24" - Major Barrel x Minor Barrel), at BBOU MS Pig Receiver (28"x24" - Major Barrel x Minor Barrel) at ELEPA MS	
3		Size - OD / WT Major Barrel	711 mm (28 inches) / 17.48mm (Reference 3)	
4		Class Rating	900#	
5		Closure Orientation	Vertical	
6		Closure Opening Direction	Open to Right	
7		Minimum Allowable Swing Angle	135°	
8		Quantity Required	2	
9	DESIGN	Design Pressure	110 barg	
10		Design Temperature	-29°C / 60°C (Min / Max)	
11		Operating Pressure	100 barg	
12		Operating Temperature		
13		Design Standard	ASME Section VIII Division 1	
14		Termination Design Standard	ASME B 31.8, ASME Section VIII Div 1.	
15		Wall Thicknesses of Major Barrel	TBA	
16		Corrosion Allowance	3.0 mm	
17	Weight of QOC	VTA (Note 9)		
18	ENVIRONMENTAL CONDITIONS	Ambient Temperature	Reference 1, 5	
19		Indoor / outdoor	Outdoor	
20		Hazardous Classification	See Note 6, 7	
21		Design Wind Speed	40 m/s	
22	OTHER REQUIREMENTS	Interlocking System E.Nwachukwu	Yes (Reference 1, Note 4)	
23		Accessories	Protective Weather Covers	
24		Testing	Reference 1, Note 5	
25		Preservation and Packing	As per Manufacturer's Standard	
26		Elastomeric Sealing	Nitrile, Viton, HNBR or any approved equivalent	
27		Painting	as per Project Painting Specification	
28		Marking	As per Project Specification Requirements	
29	MANUFACTURER'S DATA.			
30	Manufacturer's Name:	As per Approved Vendor List	Serial No.:	VTA
31	Spare Parts:	As per Project Specification		
32	Warranty requirements:	As per Manufacturer's Standard		
33	MATERIAL SPECIFICATION			
34	Part	Material Specification		
35	Major Barrel 28"	API 5L X70 PSL2	Name Plate	SS 304
36	Minor Barrel 24"	API 5L X70 PSL2	Stud bolts, external	ASTM A193 Gr B7, Hot Dip Galvanized
37	Quick Opening Barrel (QOC)	ASTM A694 F 65 / approved equal	Lifting Lugs	ASTM A516 Gr. 70
38	Gasket for QOC	VTA	Nuts, external	ASTM A194 Gr 2H, Hot Dip Galvanized
39	Davit / Hinge Bracket	SA 36 / approved equivalent		
40	FABRICATION AND INSPECTION REQUIREMENTS			
41	Construction	ASME Section VIII Division 1	Radiography	100%
42	Welding	ASME Section V	Visual Inspection	100%
43	Code Stamp	Yes	PT / MPI	100%
44	Inspection	VTA	Hydrotest	Max (ASME B31.8, ASME Sec VIII Div 1)
45	Stress Relieving	Per Code Requirement	Chemical Analysis	Per Code Requirement
46	Special Heat Treatment	Per Code Requirement	Material Certificate	EN 10204 (See Note 8)
47	REQUIREMENTS FOR VALVES INTERLOCK SYSTEM			
48	Vent Line Valves	Note 11		
49	Kicker line Valves	Note 11		
50	Drain Line Valves	Note 11		
51	Isolation Valves at Pig Trap Inlet	Note 11		

DATASHEET FOR PIG TRAP END CLOSURE		
Location:	BBOU and ELEPA Manifold Stations (MS)	Tag Numbers: BBOU-A-200, ELEP-A-200
Equipment / item:	END CLOSURE (QUICK OPENING)	
52	QUICK OPENING CLOSURE SKETCH IN OPEN POSITION (Note 10)	
		
NOTES: 1. Quick Opening Closure shall be designed as per ASME Sec VIII DIV.1 and code stamped. The Quick Opening Closure shall be free issued to the Pig Trap Vendor. 2. The quick opening closure shall be equipped with an interlock safety system that allows opening it only with the equipment being totally pressure relieved. The Safety interlock device shall be designed to make impossible to disengage the closure operating mechanism while there is pressure remaining in the equipment. 3. All moving parts of the closure shall have lubrication facilities. 4. Mechanical key interlock system as an additional safety feature shall be provided on the quick opening closure/system. Refer to the Specification for QOC and Valve Interlocks for details of interlock system 5. The Closure shall be hydrotested as part of the Pig Trap attached. 6. Hazardous Classification: Zone 1 . 7. A certificate in accordance with ATEX directives for a gas group II B, T3 shall be provided. 8. The Following material test certificates per EN 10204 shall apply: For wetted parts shall be as per certificate 3.1. For davit parts shall be as per certificate 2.2. Results of non-destructive inspections shall be documented at least by means of an acceptance test certificate 3.1. Dimension check shall be recorded in the report and certified at least by means of an acceptance test certificate 3.1. Pressure Test shall be recorded in the report and certified at least by means of an acceptance test certificate 3.2. 9. The weight of the Quick Opening Closure shall be provided by Vendor. 10. The sketch provided is for information only. VENDOR shall provide all required documentations for the QOC including the drawings. 11. As a minimum, Vent line, kicker line, drain line and isolation valves of the Pig Traps System shall be equipped with key interlock systems. These Interlocks are within Vendor Scope of Supply. Vendor shall refer to the Valve documentations referenced herein for details. REFERENCES 1. Specification for QOC and Valve Interlocks -EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-016 2. Specification Pig Traps - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011 3. Basis of Design - EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 4. Data Sheet for Piping Valves - EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-003 5. Datasheet for Mainline Valves- EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003 6. 'Process Engineering and Utility Flow Schemes (P&ID's) - EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 ABBREVIATIONS VTA - Vendor to Advise, TBC - To be Confirmed, OD-Outside Diameter, WT- Wall Thickness TBA- To be Advised		

DATASHEET FOR PIG TRAP END CLOSURE				
Location:		ELEPA Manifold Stations (MS) and AKABEL Piggng Station (PS)		Tag Numbers: ELEP-A-300, AKAB-A-100
Equipment / item:		END CLOSURE (QUICK OPENING)		
1	GENERAL	Item	Quick Opening Closure (Note 1, 2)	
2		Equipment attached	Pig Launcher (40"x 36" - Major Barrel x Minor Barrel), at ELEPA MS Pig Receiver (40"x 36" - Major Barrel x Minor Barrel) at AKABEL PS	
3		Size - OD/WT Major Barrel	1016 mm (40 inches) / 25.40mm (Reference 3)	
4		Class Rating	900#	
5		Closure Orientation	Vertical	
6		Closure Opening Direction	Open to Right	
7		Minimum Allowable Swing Angle	135°	
8		Quantity Required	2	
9	DESIGN	Design Pressure	110 barg	
10		Design Temperature	-29°C / 60°C (Min/Max)	
11		Operating Pressure	100 barg	
12		Operating Temperature		
13		Design Standard	ASME Section VIII Division 1	
14		Termination Design Standard	ASME B 31.8, ASME Section VIII Div 1.	
15		Wall Thicknesses of Major Barrel	TBA	
16		Corrosion Allowance	6.0 mm	
17		Weight of QOC	VTA (Note 9)	
18	ENVIRONMENTAL CONDITIONS	Ambient Temperature	Reference 1, 5	
19		Indoor/ outdoor	Outdoor	
20		Hazardous Classification	See Note 6, 7	
21		Design Wind Speed	40 m/s	
22	OTHER REQUIREMENTS	Interlocking Syste E.Nwachukwu	Yes (Reference 1, Note 4)	
23		Accessories	Protective Weather Covers	
24		Testing	Reference 1, Note 5	
25		Preservation and Packing	As per Manufacturer's Standard	
26		Elastomeric Sealing	Nitrile, Viton, HNBR or any approved equivalent	
27		Painting	as per Project Painting Specification	
28		Marking	As per Project Specification Requirements	
29	MANUFACTURER'S DATA.			
30	Manufacturer's Name:	As per Approved Vendor List	Serial No.:	VTA
31	Spare Parts:	As per Project Specification		
32	Warranty requirements:	As per Manufacturer's Standard		
33	MATERIAL SPECIFICATION			
34	Part	Material Specification		
35	Major Barrel 40"	API 5L X70 PSL2	Name Plate	SS 304
36	Minor Barrel 36"	API 5L X70 PSL2	Stud bolts, external	ASTM A193 Gr B7, Hot Dip Galvanized
37	Quick Opening Barrel (QOC)	ASTM A694 F 65 / approved equal	Lifting Lugs	ASTM A516 Gr. 70
38	Gasket for QOC	VTA	Nuts, external	ASTM A194 Gr 2H, Hot Dip Galvanized
39	Davit/Hinge Bracket	SA 36 / approved equivalent		
40	FABRICATION AND INSPECTION REQUIREMENTS			
41	Construction	ASME Section VIII Division 1	Radiography	100%
42	Welding	ASME Section V	Visual Inspection	100%
43	Code Stamp	Yes	PT / MPI	100%
44	Inspection	VTA	Hydrotest	Max (ASME B31.8, ASME Sec VIII Div 1)
45	Stress Relieving	Per Code Requirement	Chemical Analysis	Per Code Requirement
46	Special Heat Treatment	Per Code Requirement	Material Certificate	EN 10204 (See Note 8)
47	REQUIREMENTS FOR VALVES INTERLOCK SYSTEM			
48	Vent Line Valves	Note 11		
49	Kicker line Valves	Note 11		
50	Drain Line Valves	Note 11		
51	Isolation Valves at Pig Trap Inlet	Note 11		

DATASHEET FOR PIG TRAP END CLOSURE		
Location:	ELEPA Manifold Stations (MS) and AKABEL Pigging Station (PS)	Tag Numbers: ELEP-A-300, AKAB-A-100
Equipment / item:	END CLOSURE (QUICK OPENING)	
52	QUICK OPENING CLOSURE SKETCH IN OPEN POSITION (Note 10)	
		
NOTES: 1. Quick Opening Closure shall be designed as per ASME Sec VIII DIV.1 and code stamped. The Quick Opening Closure shall be free issued to the Pig Trap Vendor. 2. The quick opening closure shall be equipped with an interlock safety system that allows opening it only with the equipment being totally pressure relieved. The Safety interlock device shall be designed to make impossible to disengage the closure operating mechanism while there is pressure remaining in the equipment. 3. All moving parts of the closure shall have lubrication facilities. 4. Mechanical key interlock system as an additional safety feature shall be provided on the quick opening closure/system. Refer to the Specification for QOC and Valve Interlocks for details of interlock system 5. The Closure shall be hydrotested as part of the Pig Trap attached. 6. Hazardous Classification: Zone 1 . 7. A certificate in accordance with ATEX directives for a gas group II B, T3 shall be provided. 8. The Following material test certificates per EN 10204 shall apply: For wetted parts shall be as per certificate 3.1. For davit parts shall be as per certificate 2.2. Results of non-destructive inspections shall be documented at least by means of an acceptance test certificate 3.1. Dimension check shall be recorded in the report and certified at least by means of an acceptance test certificate 3.1. Pressure Test shall be recorded in the report and certified at least by means of an acceptance test certificate 3.2. 9. The weight of the Quick Opening Closure shall be provided by Vendor. 10. The sketch provided is for information only. VENDOR shall provide all required documentations for the QOC including the drawings. 11. As a minimum, Vent line, kicker line, drain line and isolation valves of the Pig Traps System shall be equipped with key interlock systems. These Interlocks are within Vendor Scope of Supply. Vendor shall refer to the Valve documentations referenced herein for details. REFERENCES 1. Specification for QOC and Valve Interlocks -EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-016 2. Specification for Pig Traps - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011 3. Basis of Design - EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 4. Data Sheet for Piping Valves - EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-003 5. Datasheet for Mainline Valves- EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003 6. 'Process Engineering and Utility Flow Schemes (P&ID's) - EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 ABBREVIATIONS VTA - Vendor to Advise, TBC - To be Confirmed, OD-Outside Diameter, WT- Wall Thickness TBA- To be Advised		